BiomedCLIP: BiomedCLIP-PubMedBERT_256-vit_base_patch16_224

Problem

-manual processing of structured biomedical data is costly and difficult to scale
-many existing pretraining datasets are private/small
-lack of Diversity (existing datasets focus heavily on chest X-rays)
-general-domain models (original CLIP) perform poorly on medical tasks
-standard models use a context limit of 77 tokens but biomedical captions in scientific papers are much longer

Evaluation

-cross-modal retrieval: test the model's ability to retrieve the correct image given a text caption (text2img or img2txt), held-out test set from PMC-15M containing 725,739 image-caption pairs, precomputed image and text embeddings and performed an approximate nearest neighbor search using cosine similarity, as metric they use Recall@k which measures if the correct match was within the top k results
-image classification: covered zero-shot, few-shot and full-shot (linear probing) scenarios using datasets of PCam (binary classification of metastatic tumor tissue in lymph node histopathology), LC25000 (Lung & Colon: classification of benign vs malignant lung and colon tissues), TCGA-TIL (detection of tumor-infiltrating lymphocytes in whole-slide images), RSNA (detection of pneumonia in chest radiographs), as metrics they used accuracy for most datasets and AUROC for TCGA-TIL (due to class imbalance), for zero-shot tasks they used specific text prompts ("a photo of {}; pneumonia {} presented in image") to assist the model without any further training
-visual question answering (VQA): the model was evaluated on its ability to answer clinical questions about radiology images, they used VQA-RAD (3,515 question-answer pairs) and SLAKE datasets (over 7,000 pairs), they utilized the METER framework which treats VQA as a classification task by fusing image and text encodings to predict a final answer, as metrics they used accuracy (string-level correctness) and token-level F1 (using GPT-2 for token matching)

Dataset

-create PMC-15M dataset
-contains 15 million biomedical image-text pairs collected from 4.4 million scientific articles in PubMed Central (PMC)
-PMC-15M covers 30 major image types including radiography (MRI, CT, X-ray), digital pathology, microscopy, non-clinical illustrations (graphs, charts)
-Download full-text articles from PMC-OA then extract figure-caption pairs from the articles
-use image and their captions (no extra text and include plots)
-PMC-Fine-Grained-46M: separate and more complex version that includes extra text references

Training ptotocol

-domain-specific adaptation of the CLIP framework, biomedCLIP was pretrained on PMC-15M dataset and instead of training from scratch the text encoder is initialized with PubMedBERT which used wordPiece tokenizer and contect length of 256 tokens, they utilized ViT architecture, they initialized ImageNet-pretrained weights after testing random initialization, the images are processed at resolution of 224*224
-Epochs = 32
-Optimizer: AdamW (decouples weight decay (L2 regularization) from the gradient update)
-Learning Rate = 5.0e-4
-Weight Decay = 0.2
-Batch Size = 4k (4,000 pairs)

Future work

-they intend to extract in-line references (citances) from scientific papers and pair them with figures to provide additional training data
-the team plans to leverage the PMC-Fine-Grained-46M dataset to enhance pretraining which involves splitting composite figures into individual panels to allow the model to better understand detailed visual grounding
-they have only used the ViT-B vision encoder but in the future they want to explore larger Vision Transformer models such as ViT-L, ViT-H and ViT-G
-higher image resolution
-they're interested in applying their multimodal methodology to other types of biomedical data beyond just images and text such as gene expression and molecular sequence data
-authors hope to explore more complex applications including multimodal generation, advanced image search and precision health through multimodal fusion

Proposed approach: BiomedCLIP

-specialized VLM
-CLIP (Contrastive Language-Image Pretraining), a modeling approach that trains an image encoder and a text encoder simultaneously to project both types of data into a shared embedding space, it's optimized to recognize which image belongs with which text by maximizing the cosine similarity between matching (positive) pairs and minimizing it for non-matching (negative) pairs through InfoNCE loss, created by openAI
-biomedical foundation model
-designed to represent images and text in a shared embedding space
-domain-specific text encoding: replace the general-domain GPT-2 model used in standard CLIP with PubMedBERT, swapped Byte-Pair Encoding (BPE) for a WordPiece tokenizer, text context size was increased from 77 to 256 tokens
-specialized vision encoding: utilize ViT architecture (ViT-B/16 model: base model + image is split into 16×16 pixel patches) to process images, it's initialized with ImageNet-pretrained weights
-the core of the approach is pretraining the model on PMC-15M
-to handle the massive scale of the data and compute requirements:
model uses InfoNCE loss to maximize the cosine similarity between matching image-text pairs while minimizing it for non-matching pairs
utilize patch dropout strategy to enhance performance and sharding contrastive loss to reduce memory consumption during training on multiple GPUs
batch size of 4k (they tried even 64k)

Results

-BiomedCLIP achieved new state-of-the-art results
-Cross-Modal Retrieval:
accuracy: out of 700k+ candidates, the top-1 results were correct over 56% of the time, and the top-5 results were correct over 77% of the time
-recall: 88.80% for lung opacity and 95.04% for atelectasis

Limitation

-currently, the model only uses figure captions for training (no extra text)
-roughly half of the images in the dataset are composite figures (multiple images in one file), while they developed a separate Fine-Grained dataset to address this but the current BiomedCLIP was not trained on split sub-figures
-due to computational constraints the largest vision encoder used was ViT-B which is considered relatively small compared to larger architectures like ViT-L, ViT-H, ViT-G
-the model utilizes a relatively low input resolution (224*224 pixels) while over 75% of the images in the PMC-15M dataset are larger than 336*336 pixels
-the current research is restricted to images and text, authors noted it would be beneficial to apply this methodology to other biomedical data types such as gene expression and molecular sequence data
-when used for privacy-preserving analysis of clinical data, the model can be limited by the nature of scientific writing because scientific captions may not always describe conditions as comprehensively as clinical reports, for example the model showed lower recall for "Cardiomegaly" compared to other conditions because it was less frequently or detailedly mentioned in the training captions